

ELEC-1107 – Project Expectation, Clarification and Rubric

(Project: Powered Speaker and Mobile Device Charging Station)

The project will be a phased approach to applying into practice many of the concepts studied in this electronic devices course. In order to study some “real life” circuit applications, the students will perform design validation and component selection calculations for the project designs. The project involves building a 5 VDC, 3 Ampere USB (Universal Serial Bus) power supply for charging mobile devices. It will also incorporate a powered audio amplifier with a 1/8” audio jack input with a built in speaker. Finally, it will also include some basic safety and monitoring circuits.

The project will be split into 3 sections with a total weight of 30% of the final course grade and will be graded as outlined in the rubric and in accordance with the CVLOs and EESs as outlined in our program mapping and listed in the ministry guidelines. The project will be graded in week 14 with the following grading breakdown:

Section 1 of the project will be the finished product and will be worth 1/3 of the project grade (10% of your final grade). The finished product will be graded on functionality, layout, fabrication techniques and professionalism as outlined in the rubric. Since the students have successfully completed a skills (fabrication and soldering) course, a portion of the fabrication grade will involve soldering all the components on a perforated prototype board.

Section 2 of the project will be an evaluation of the demonstration of the performance characteristic measurements of the project and will be worth 1/3 of the project grade (10% of your final grade). For example, some of the performance characteristic measurements for the power supply portion might include line and load regulation, ripple and ripple factor, etc. The demonstration of the performance characteristic measurements will be graded as outlined in the rubric and in accordance with the CVLOs and EESs as outlined in our program mapping and listed in the ministry guidelines.

The third and final portion (Section 3) of the project will be in the written lab report and is worth the remaining 1/3 of the grade (10% of your final grade). The lab report will be written with the guidelines as described below and graded against the rubric below. The written lab report portion will be graded as outlined in the rubric and in accordance with the CVLOs and EESs as outlined in our program mapping and listed in the ministry guidelines.

Section 2 – Design Validation and Performance Characteristics

In this part of the project, you will be performing nominal calculations so that you know what to expect when performing measurements. The following are some examples of the data your paper must have. Be sure to include device pin-outs and relevant data sheet information.

When performing measurements using a single phase transformer as the source voltage, please use the center tap (or one half of the full secondary) voltage as the source (E_{S1}). Set the oscilloscope's input channels to DC coupling with the "BW Limit" (Bandwidth Limit) on. Be sure to maximize the waveform on the screen. Be sure to trigger on channel 1 and have channel 1 cross at (0,0). Use the oscilloscope function to measure to display all the required or appropriate data. Be sure to take photos of the oscilloscope screen to use in your report later. When required, use the "Run/Stop" button to assist in holding the image in one spot to assist in taking the photo. When taking photos, make sure the image is straight/square so that when you crop the images and add a solid line around them, they'll look professional.

You should also include a schematic diagram of each of the section circuits as well as photos of your board for each of the sections. When taking photos of your board, be sure they are in focus (and not blurry) as well as to be sure the lighting is appropriate. Make sure you use a white background under your board so that after you put a solid line border around the image, your photos look professional.

Once you are satisfied with your results, please document how much time you spent on the design calculations/simulations, build, troubleshooting, design validation and documentation for this sections. Have the instructor/lab technician sign and date your rubric. Your rubric must be signed and dated for every section before moving on to the next section of the project. Do not move ahead in the build process without having the previous section signed-off by the instructor/lab technician or you will be asked to remove the unapproved section components.

Section A – Voltage Rectifier Circuit

Please perform and include all the following calculations with assumptions:

Assumption:

$E_{\text{Primary(RMS)}} = 120 \text{ V}$, $a = 4.6$ (4.6 : 1), $V_{\text{Diode}} = 0.7 \text{ V}$, $I_{\text{T-DC(Max)}} = 1.0 \text{ A}$, $I_{\text{T-AC(Pk)}} = 250 \text{ mA}$,

Calculations:

$E_{\text{CT-Secondary(RMS)}}$, $E_{\text{CT-Secondary (Pk)}}$, $E_{\text{CT-Secondary (P-P)}}$, $V_{\text{Rectified(Pk)}}$, V_{PIV} , $V_{\text{Ripple(P-P)}}$, $V_{\text{Ripple(RMS)}}$, V_{DC} , $r_{\text{Ripple-Factor(\%)}}$, $R_{\text{Load-Voltage-Regulation(\%)}}$, $I_{\text{Effective}}$ and P_{T} .

Documented Information:

When performing measurements for this section, please use the center tap (or one half of the full secondary) voltage of the single phase transformer as the source voltage. Set the oscilloscope's Channel 1 to 5 V/Div. and with DC coupling. Be sure to maximize the waveform on the screen. Use the oscilloscope's measure function to display $V_{\text{(P-P)}}$, V_{RMS} , Period, V_{Max} and V_{Min} . The frequency should already displayed on the bottom/left of the screen. Use channel 1 to measure the rectifier's input and output signals. Be sure to take a photo of the oscilloscope screen to use in your report later. Use the "Run/Stop" button to assist in holding the image in one spot to assist in taking the photo.

Measurements/Data:

$E_{\text{Primary(RMS)}}$, $E_{\text{Secondary(RMS)}}$, $E_{\text{Secondary (Pk)}}$, $E_{\text{Secondary (P-P)}}$, a , V_{Diode} , $V_{\text{Rectified(Pk)}}$, $V_{\text{Ripple(P-P)}}$, $V_{\text{Ripple(RMS)}}$, V_{DC} , R_5 , C_1 , $V_{\text{F(LED2)}}$,

Simulation/Data:

$E_{\text{Primary(RMS)}}$, $E_{\text{Secondary(RMS)}}$, $V_{\text{Rectified(Pk)}}$ and the MultiSim oscilloscope output waveform

Relevant Data Sheet Information:

$I_{\text{Average(RMS)}}$, $V_{\text{Forward-Voltage}}$, I_{Reverse} , $V_{\text{PIV-Maximum}}$, $P_{\text{D-Maximum}}$.

Please also include a diode and LED pin-out and schematic diagram.

Section B – Under-Voltage Detector Circuit

Please perform and include all the following calculations with assumptions:

Assumption:

$V_{CE(\text{Minimum})} = 1.8 \text{ V}$, $I_{C(\text{Minimum})} = 268 \mu\text{A}$, $\beta_{DC(\text{Minimum})} = 178.667 \text{ mA/mA}$, $V_{BE} \approx 610 \text{ mV}$, $V_{Z(\text{Minimum})} = 5.91 \text{ V}$, $V_{CC(\text{Maximum})} = 17.75 V_{DC}$, $V_{CE(\text{Saturation})} \approx 50 \text{ mV}$, $V_{Z(\text{Maximum})} = 6.10 \text{ V}$,

Calculations: (at the assumed minimum “Enable” point of the voltage regulators ON/OFF pin)

$V_{RC(\text{Minimum})}$, $I_B(\text{Minimum})$, $I_{R2(\text{Minimum})}$, $I_{R1(\text{Minimum})}$, $V_{R1(\text{Minimum})}$, $I_E(\text{Minimum})$, $V_{CC(\text{Minimum})}$, $V_{RC(\text{Maximum})}$, $I_C(\text{Maximum})$, $V_{R1(\text{Maximum})}$, $I_{R1(\text{Maximum})}$, $I_B(\text{Maximum})$, $I_E(\text{Maximum})$, $\beta_{DC(\text{Maximum})}$, $P_D(\text{Maximum})$, $P_T(\text{Maximum})$.

Documented Information:

When performing measurements for this section, please use your DMM and the DC power supply (as the source voltage). Set the power supply to 0 VDC and the DMM to measure V_{CE} . Using the coarse and then the fine adjustment dials, increase the DC source voltage until V_{CE} is 1.8 V and then measure E_{S1} . This is the “minimum” the source voltage ($E_{S1(\text{Minimum})}$) that the source voltage can drop to before the voltage regulator shuts down. Then, measure the required voltages while the DC power supply is set to 4 V_{DC} and then to 15 V_{DC} .

Measurements/Data:

$V_{CC(\text{Minimum})}$, $V_{R2(@VCC=4V)}$, $V_{R1(@VCC=4V)}$, $V_Z(@VCC=4V)$, $V_{CE(@VCC=4V)}$, $V_{RC(@VCC=4V)}$, $V_{R2(@VCC=15V)}$, $V_{R1(@VCC=15V)}$, $V_Z(@VCC=15V)$, $V_{CE(@VCC=15V)}$, $V_{RC(@VCC=15V)}$

Simulation/Data:

$V_{R2(@VCC=4V)}$, $V_{R1(@VCC=4V)}$, $V_Z(@VCC=4V)$, $V_{CE(@VCC=4V)}$, $V_{RC(@VCC=4V)}$, $V_{R2(@VCC=15V)}$, $V_{R1(@VCC=15V)}$, $V_Z(@VCC=15V)$, $V_{CE(@VCC=15V)}$, $V_{RC(@VCC=15V)}$

Relevant Data Sheet Information:

$\beta_{DC(@10mA)}$, $V_{CE(\text{Saturation}@10mA)}$, $V_{BE(\text{Saturation}@1mA)}$, $I_C(\text{Maximum})$, $P_D\text{-Maximum}$.

Please include a pin-out diagram and schematic diagram of the BJT.

Section C – 5 VDC Regulation Circuit

Please perform and include all the following calculations with assumptions:

Assumption:

$$V_{DC(\text{Unloaded})} = 5.2 V_{DC}, V_{F(\text{LED1})} = 1.8 V_{DC},$$

Calculations:

$$V_{R4}, I_{R4}, I_{\text{LED1}}, P_{R4}, P_{D(\text{LED1})}.$$

Documented Information:

When performing measurements for this section, please use the center tap (or one half of the full secondary) voltage of the single phase transformer as the source voltage. Set the oscilloscope's Channel 1 to 5 V/Div. and with DC coupling and move the channel 1 signal trace 2 divisions below center. Set the time base to 5 μ s/Div. Use the oscilloscope's measure function to display $V_{(\text{Mean})}$, $V_{(\text{Max})}$, $V_{(\text{Min})}$, Period and frequency. Use channel 1 to display the voltage across the Schottky diode (D1) and the regulators output voltage. Be sure to take photos of the oscilloscope screen to use in your report later.

Measurements/Data:

$$V_{\text{Input(Average-DC)}}, V_{\text{Output(Average-DC)}}, T_{\text{Oscillation}}, f_{\text{Oscillation}}, V_{R4}, V_{f(\text{LED1})},$$

Simulation/Data:

A simulation is not required for this section as the LM2576 is not in the MultiSim Library.

Relevant Data Sheet Information:

$$f_{\text{Oscillation(Minimum)}}, f_{\text{Oscillation(Maximum)}}, f_{\text{Oscillation(Typical)}}, V_{\text{In(Maximum)}}, V_{\text{Control(ON-Minimum)}}, V_{\text{Control (OFF-Maximum)}}, \\ V_{\text{Output@0.5A<=I-Load<=3A(Minimum)}}, V_{\text{Output@0.5A<=I-Load<=3A(Maximum)}}, \eta(\text{Percent-Efficiency}),$$

Please include a pin-out diagram of the voltage regulator IC and schematic diagram.

Section D – Start Up Delay Circuit

Please perform and include all the following calculations with assumptions:

Assumption:

$$V_{DC(\text{Rectified})} = 17.75 \text{ V}, V_{BE(\text{Darlington})} \approx 2 \cdot (750 \text{ mV}) = 1.5 \text{ V}, V_{CE(\text{Darlington})} \approx 2V_{BE} + 2V_{CE(\text{Sat})} = 2 \cdot (750 \text{ mV}) + 2 \cdot (200 \text{ mV}) = 900 \text{ mV}, V_{DC(\text{Unloaded})} = 5.2 \text{ V}$$

Calculations:

$$\tau, V_{C3(\text{Max}@t \geq 5\tau)}, V_{R8(\text{Minimum-to-Switch})}, V_{R7(\text{Minimum-to-Switch})}, V_{C3(\text{Minimum-to-Switch})}, t_{(\text{Switch})}, V_{\text{Relay}},$$

Documented Information:

When performing measurements for this section, please use the center tap (or one half of the full secondary) voltage of the single phase transformer as the source voltage. With capacitor C_3 fully discharged and a stopwatch reset, turn on the transformer and time how long it takes for the relay coil to be energized ($t_{(\text{Switch})}$). Set the oscilloscope's Channel 1 to 5 V/Div. and with DC coupling and move the channel 1 signal trace 2 divisions above center. Set Channel 2 to 5 V/Div. and with DC coupling and move the channel 2 signal trace 2 divisions below center. Set the time base to 25 ms/Div. Use the oscilloscope's measure function to display $V_{CH1(\text{Mean})}$ and $V_{CH2(\text{Mean})}$. Use channel 1 to display V_{C3} and channel 2 to display V_{R8} . Be sure to take photos of the oscilloscope screen to use in your report later. Once that is done, use your DMM to measure $V_{BE(\text{Darlington})}$, $V_{CE(\text{Darlington})}$ and V_{Relay} .

Measurements/Data:

$$t_{(\text{Switch})}, V_{BE(\text{Darlington})}, V_{CE(\text{Darlington})}, V_{\text{Relay}}$$

Simulation/Data:

$$V_{BE(\text{Darlington})}, V_{CE(\text{Darlington})}, t_{(\text{Switch})}, V_{C3(\text{Max}@t \geq 5\tau)}, V_{R8(@t\text{-Switch})}, V_{R7(@t\text{-Switch})}, V_{C3(@t\text{-Switch})}, V_{\text{Relay}}$$

Relevant Data Sheet Information:

$$I_{(\text{Relay-Contact})}, V_{\text{Relay(On-Minimum)}}, V_{\text{Relay(Off-Minimum)}}, N_{\text{Expected-Mechanical-Life(Cycles)}}, N_{\text{Expected-Electrical-Life(Cycles)}}$$

Please include a pin-out diagram of the Darlington pair and the relay and schematic diagram.

Section E – 5 VDC USB Charging Circuit

Please perform and include all the following calculations with assumptions:

Assumption:

$$V_{DC(\text{Unloaded})} = 5.2 \text{ V}$$

Calculations:

$$V_{R-DP1}, V_{R-DP2(D+)}, V_{R-DN1}, V_{R-DN2(D-)}$$

Documented Information:

When performing measurements for this section, please use the center tap (or one half of the full secondary) voltage of the single phase transformer as the source voltage. Use your DMM to take voltage measurements across all the resistors and the voltage between the data pins (D+ and D-).

Measurements/Data:

$$V_{R-DP1}, V_{R-DP2(D+)}, V_{R-DN1}, V_{R-DN2(D-)}, \text{ voltage at D+ with respect to D-}$$

Simulation/Data:

$$V_{R-DP1}, V_{R-DP2(D+)}, V_{R-DN1}, V_{R-DN2(D-)}, \text{ voltage at D+ with respect to D-}$$

Relevant Data Sheet Information:

Please include a pin-out diagram of the USB board header pins and schematic diagram.

Section F – Audio Amplifier Circuit

Please perform and include all the following calculations with assumptions:

Assumption:

$V_{DC(Rectified)} = 17.75 \text{ V}$, $A_v = 50 \text{ V/V}$, $V_{In(Test)} = 95 \text{ mV}_{RMS}$, $f_{In(Test)} = 1 \text{ kHz}$.

Calculations:

$V_{In-Test(Pk-Pk)}$, $V_{In-Test(Pk)}$, A_v , $V_{Out-Test(Pk-Pk)}$, $V_{Out-Test(Pk)}$, $V_{Out-Test(RMS)}$, $P_{D(Max)}$

Documented Information:

When performing measurements for this section, please use the center tap (or one half of the full secondary) voltage of the single phase transformer as the source voltage. Set the function generator to produce an sinusoidal waveform at 1 kHz. Set the oscilloscope's Channel 1 to 50 mV/Div. and with DC coupling 2 divisions above center. Set Channel 2 to 5 V/Div. and with DC coupling 2 divisions below center. Set the time base to 250 μ s/Div. Use the measure function to display $V_{CH1(Pk-Pk)}$, $V_{CH1(RMS)}$, $V_{CH2(Pk-Pk)}$, $V_{CH2(RMS)}$ and $T_{(Period)}$. The frequency should already displayed on the bottom/left of the screen. Connect the function generator and channel 1 of the oscilloscope to the input and a 10 Ω resistor and channel 2 across the output. Increase the input amplitude until just before the output signal becomes distorted. Be careful as the IC will get hot fast. Be sure to take photos of the oscilloscope screen for your report.

Measurements/Data:

$V_{In-Test(Pk-Pk)}$, $V_{In-Test(Pk)}$, $V_{In-Test(RMS)}$, $V_{Out-Test(Pk-Pk)}$, $V_{Out(Pk)}$, $V_{Out(RMS)}$, A_v , $I_{Out(RMS)}$, $I_{Out(Pk-Pk)}$, $P_{D(Output)}$

Simulation/Data:

A simulation is not required for this section as the LM380 is not in the MultiSim Library.

Relevant Data Sheet Information:

$P_{Out(RMS)}$, $A_v(Typical)$, $V_{Out(Typical)}$, $Z_{In(Typical)}$, $THD(Typical)$, $V_{S(Minimum)}$, $V_{S(Maximum)}$, $I_{Short-Circuit(RMS)}$, A_v

Please include a pin-out diagram of the audio amplifier IC and of a 1/8" phone mono and a 1/8" phone stereo jack (or aux plug). Also, please include a schematic diagram.